

Vision: From Algorithm to First Principles

The AGM framework is not the destination—it is proof that the destination exists.

The Trajectory

This proposal presents a specific mathematical framework—complete elliptic integrals computed via AGM—as the computational kernel for quantitative LLMs. This specificity is deliberate but temporary. The AGM framework serves as auxiliary wheels: a proven implementation demonstrating that physics-informed quantitative reasoning is architecturally achievable, computationally efficient, and empirically valid.

The ultimate vision transcends any particular algorithm. It rests on first principles that govern all physical systems amenable to quantitative reasoning.

First Principles: The Invariant Core

Regardless of specific mathematical machinery, any successful quantitative reasoning system must satisfy five foundational requirements:

1. Conservation as Hard Constraint

Physical laws (mass, energy, momentum, charge) cannot be violated—not as soft penalties minimized during training, but as architectural impossibilities enforced by construction. The AGM framework achieves this through transfer function passivity. Future algorithms might use differential geometry, Hamiltonian mechanics, or symplectic integrators. The principle remains: conservation is structurally guaranteed, not statistically approximated.

2. Dimensional Consistency and Scale Invariance

Physical systems admit dimensional analysis—relationships between quantities must be dimensionally consistent, and physics is invariant under appropriate scalings. The AGM framework's ξ parameter achieves scale-free representation through logarithmic normalization. Future systems might use renormalization group theory, similarity solutions, or Buckingham π -theorem. The principle remains: extract dimensionless invariants that capture essential physics independent of arbitrary units.

3. Causality and Temporal Ordering

Physical processes have causal structure—outputs depend on past inputs, not future ones. The AGM framework encodes this in transfer function pole-zero topology (all poles in left half-plane). Future approaches might use directed acyclic graphs, discrete event systems, or temporal logic. The principle remains: time's arrow is respected at the computational level, not merely in post-processing.

4. Topology of Solution Manifolds

Physical systems occupy manifolds in state space with characteristic topological features—fixed points, limit cycles, bifurcations, separatrices. The AGM framework captures this through discrete topological invariants (S , the pole-zero parity). Future methods might use persistent homology, Morse theory, or dynamical systems theory.

The principle remains: discrete topological features provide robust, noise-tolerant descriptors of system state.

5. Multi-Domain Synchronization

Real systems coordinate multiple conservation laws simultaneously—satisfying mass balance while respecting energy constraints while maintaining momentum closure. The AGM framework handles this through concurrent evaluation of domain-specific transfer functions. Future architectures might use constraint satisfaction networks, multi-port Hamiltonian systems, or concurrent state machines. The principle remains: constraint satisfaction in one domain must synchronize with others through event propagation, not sequential checking.

Hardware-Algorithm Co-Evolution

The Event Processing Unit architecture is deliberately agnostic to specific algorithms. EPU provides primitive operations for any constraint-driven computation:

Binary constraint states: Conservation law satisfied (1) or violated (0)

Event propagation: When constraint C_1 is satisfied, notify all computations waiting on C_1

Multi-domain coordination: Concurrent state machines for mass, energy, momentum

Provenance tracking: Which events triggered which state transitions

Whether the algorithm uses elliptic integrals, Lie group methods, or variational principles is irrelevant to the EPU. The hardware provides the substrate; the algorithm exploits it. This separation enables algorithm innovation without hardware redesign.

The Development Pathway

Phase I (Years 1-2): Learn from the Specific

Implement and validate the AGM framework. Understand why it works: not because elliptic integrals are magical, but because they efficiently encode conservation, causality, and topology. Extract the design patterns that make it successful.

Phase II (Years 3-4): Abstract to the General

Develop alternative algorithms satisfying the same first principles—Hamiltonian neural networks, port-Hamiltonian systems, symplectic integrators, physics-informed neural operators. Benchmark against AGM baseline. Identify which principles are algorithm-independent and which are implementation artifacts.

Phase III (Years 5+): Systematize the Principles

Formalize a mathematical theory of physics-informed quantitative reasoning. Prove theorems about what classes of systems admit efficient constraint-respecting computation. Design algorithms directly from first principles rather than adapting existing mathematical tools. The AGM framework becomes a historical footnote—important for proving feasibility, superseded by principled design.

The End State

In ten years, quantitative LLMs should no more depend on specific special functions than modern GPUs depend on specific matrix algorithms. GPUs provide primitive operations (FMA, memory bandwidth) that enable efficient linear algebra; the specific algorithm (BLAS, cuBLAS, vendor libraries) is replaceable.

Similarly, quantitative LLMs should provide primitive operations (conservation enforcement, dimensional analysis, causal ordering, topology extraction, multi-domain synchronization) that enable efficient physics-informed reasoning. The specific algorithm—whether elliptic integrals, Hamiltonian mechanics, or methods not yet invented—becomes an implementation detail.

The measure of success is not whether the AGM framework dominates, but whether it becomes unnecessary—replaced by a principled science of quantitative reasoning that it helped establish.

First principles are eternal. Algorithms are scaffolding.